

CONSTRUCTION OS

PRD_Procurement_Module

Procurement, Suppliers, Sourcing and Purchase Control

Product definition

The enterprise purchasing engine for purchase demand, supplier qualification, sourcing, quotation comparison, purchase orders, delivery commitments, receiving context, supplier performance and complete procurement traceability. Procurement owns purchasing workflows while connected modules remain authoritative for stock, accounting, legal agreements, assets, projects, documents and tasks.

Field	Value
Document type	Full Product Requirements Document
Status	Implementation-ready functional specification
Version	1.0
Date	5 August 2026
Priority	Core ERP / operational module
Audience	Product, UX/UI, Frontend, Backend, Data, QA, Security and DevOps teams
Approval architecture	Integration-ready only; the central Approvals and Workflow Engine is intentionally deferred to the final platform PRD

Design direction: premium, precise, auditable and construction-aware.

1. Document Control

Field	Value
Document	PRD_Procurement_Module
Primary routes	/procurement, /procurement/suppliers, /procurement/requests, /procurement/rfqs, /procurement/orders
Primary pages	Procurement Dashboard, Procurement Workspace, Supplier Page, Purchase Request Page, Procurement Package Page, RFQ Page, Comparison Workspace, Purchase Order Page, Delivery Page, Reports and Settings
Connected modules	Group, Company, Departments, Teams, Projects, Units, Inventory, Assets, Finance, Contracts, Documents, Tasks, Calendar, Notifications and future Approvals
Access model	Full capability first for authenticated users; all actions must call permission hooks so later role, record and field restrictions can be introduced without redesign
Platform Configuration	Mandatory at module, page, section, widget, feature, field group, workflow checkpoint, report and action level
Core source-of-truth rule	Procurement owns supplier purchasing context and the purchasing lifecycle. Inventory owns stock and posted inventory movements. Finance owns budgets, AP invoices, payments and accounting. Contracts owns signed legal agreements. Assets owns long-lived asset lifecycle. Documents owns files and revisions.
External integrations	Not required for initial implementation; architecture remains integration-ready
Approvals dependency	This PRD identifies authorization checkpoints and payloads only. It does not define the central routing, delegation or escalation engine.

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3. Executive Summary

The Procurement Module is the platform-wide source of truth for supplier purchasing context and the complete purchasing lifecycle. It turns project, warehouse, asset, maintenance, department and company demand into controlled sourcing events, comparable offers, purchase commitments, delivery plans and measurable supplier outcomes.

The module must support simple office purchasing and complex construction procurement without creating separate data models. One Purchase Request may contain stock items, assets, services, rentals or subcontract work. One Procurement Package may group multiple requests, scopes, suppliers, RFQs, orders, deliveries and contracts around a project package such as Structure, Facade, MEP, Fit-Out or Site Logistics.

Core product statement

Procurement is not a purchase-order list. It is the controlled commercial operating layer between demand, suppliers, commitments, delivery and the authoritative records held by Inventory, Finance, Contracts, Assets and Projects.

- **Universal but construction-aware.** Support materials, equipment, assets, services, subcontract scopes, rentals, recurring orders and project packages.
- **Multi-company from first release.** Every record has tenant, owning company, optional project and commercial scope.
- **Traceable by design.** Every decision, revision, comparison, award, delivery exception and supplier status change is auditable.
- **No duplicated ownership.** The module links to stock receipts, financial postings, contracts and documents rather than reproducing them.
- **Permission-ready.** The current allow-all phase still requires capability checks, tenant isolation and field-level policy hooks.
- **Workflow-ready, not workflow-owned.** Approval checkpoints are exposed for a future platform engine; the engine itself is explicitly outside this PRD.

4. Product Vision, Goals and Non-Goals

4.1 Product Vision

- Provide one enterprise procurement workspace across group companies, departments, projects, sites and warehouses.
- Convert operational demand into competitive, controlled and measurable purchasing outcomes.
- Make total evaluated cost, delivery risk, quality, compliance and supplier history visible before award.
- Protect commercial history through immutable issued records and revision-based change control.
- Allow management to understand requested, quoted, committed, received, invoiced and paid values without confusing source ownership.
- Give construction teams a Procurement Package that mirrors project work packages and delivery sequences.
- Provide premium desktop workflows and mobile field actions without reducing auditability.

4.2 Product Goals

- Know what is being requested, by whom, for which company/project/location and when it is needed.
- Know which suppliers were invited, what they offered, how offers were normalized and why an award was recommended.
- Prevent duplicate buying, uncontrolled supplier use, silent PO changes and untraceable delivery variances.
- Connect purchase commitments to approved Finance budgets and project cost dimensions.
- Generate Inventory receiving context and Asset draft records without owning their lifecycle.
- Measure supplier delivery, quality, commercial and compliance performance using source-linked evidence.
- Scale from one buyer to a multi-company procurement organization with millions of line and activity records.

4.3 Non-Goals

- Inventory quantity, warehouse balances, batch/serial stock history and posted stock movements.
- General ledger, accounts payable posting, bank payments, tax filing or statutory accounting.
- Contract authoring, signed legal terms, amendments and legal obligation management.
- Asset depreciation, assignment, maintenance and disposal lifecycle.
- Final Approvals and Workflow Engine, including routing, delegation, escalation and organization-wide decision policy.
- Supplier self-service portal, e-auctions, EDI, marketplace integrations, live courier tracking, OCR and AI in the first release.
- Jurisdiction-specific procurement law or public-tender compliance advice.

5. Architectural Ownership and Domain Boundaries

Domain	System of record	Procurement behavior
Supplier purchasing profile	Procurement	Owns qualification status, categories, commercial terms, sourcing participation, performance, risk flags and procurement documents.
Shared legal/company identity	Company / master data layer	Links supplier to a reusable business-party identity where available; does not duplicate group/company records.
Purchase demand, RFQ, quote, comparison, award and PO	Procurement	Owns lifecycle, line-level commercial data, delivery commitments and revision history.
Product master and stock	Inventory	Reads product/SKU/unit data; sends receiving expectations; links exact Inventory Receipt and movement records.
Budgets, commitments, invoices, payments and accounting	Finance	Reads approved budgets and actual finance states; emits commitment-ready events; displays source-linked values.
Signed purchase/subcontract/framework agreement	Contracts	Procurement may initiate contract creation and call-off use; Contracts owns legal text, parties and effective agreement.
Long-lived asset lifecycle	Assets	Receiving may create a draft Asset; Assets owns activation, assignment, maintenance, depreciation context and disposal.
Project hierarchy and work context	Projects	Links company, project, structure, floor, unit, phase, milestone, work package, cost code and site.
Files and revisions	Documents	All attachments are Document records with Document Passport, revisions, shortcuts and Transit promotion.
Execution work	Tasks	Procurement can create or link participant-based tasks; Tasks owns task state and collaboration.
Decisions and approval routing	Future Approvals Engine	Procurement defines checkpoints and request payloads only; the future engine owns routing and decisions.

PROC-001 No procurement page may expand a user's access to confidential Finance, Contract, Document, Project or Supplier data merely because a linked summary is visible.

PROC-002 Every cross-module link must store the authoritative record ID, source module, version where relevant and a correlation ID for end-to-end traceability.

PROC-003 Issued commercial records must not be edited in place. Corrections and changes must use a revision, cancellation, replacement or reversal-style record.

6. Users and Core Use Cases

User archetype	Primary use cases
Requester	Create a purchase request, describe need, attach specifications, identify project/location and track fulfillment.
Buyer / Procurement Specialist	Source suppliers, issue RFQs, capture quotations, normalize offers, negotiate, recommend and issue purchase orders.
Procurement Manager	Manage packages, suppliers, exceptions, commercial risk, spend, team workload and authorization checkpoints.
Project Manager / Site Manager	Create project demand, review technical fit, confirm delivery priorities and monitor project procurement.

User archetype	Primary use cases
Technical Evaluator	Evaluate compliance with drawings, specifications, quantities, samples, method statements and technical scope.
Warehouse / Inventory User	View expected deliveries, perform authoritative inventory receiving and link receipt/rejection results.
Asset Manager	Review asset-related purchases and continue lifecycle from linked draft assets.
Finance User	Validate budgets, view commitments, handle AP invoices, matching, payments and accounting.
Contract / Legal User	Create and manage formal agreements after award where required.
Executive / Auditor	Review spend, supplier concentration, savings, delays, exceptions and immutable activity history.

7. Information Architecture

Navigation area	Pages / workspaces
Overview	Procurement Dashboard, My Work, Exceptions, Recent Activity
Demand	Purchase Requests, Replenishment Recommendations, Emergency Requests
Packages	Procurement Packages, Package Page, Package Timeline, Package Budget and Deliveries
Sourcing	RFQs, RFQ Page, Supplier Invitations, Quotations, Comparison Workspace, Negotiation Rounds, Award Recommendation
Ordering	Purchase Orders, PO Page, PO Revisions, Call-Off Orders, Framework Usage
Delivery	Delivery Schedule, Delivery Page, Expected Receipts, Service Receipts, Returns, Claims and Shortages
Suppliers	Supplier Directory, Supplier Page, Qualification, Categories, Documents, Scorecards, Risk Flags
Analytics	Spend, Savings, Cycle Time, Delivery, Quality, Supplier Concentration, Off-Contract and Project Procurement
Administration	Procurement Settings, Numbering, Categories, Evaluation Templates, Tolerances, Statuses, Platform Configuration

7.1 Global Record Rules

- Every primary record receives a permanent UUID and a configurable human-readable code.
- Every record stores tenant, owning company, creator, current status, created/updated timestamps and immutable activity references.
- Project context is optional globally but mandatory when company policy or request category requires it.
- Lines, not only headers, may carry project, work package, cost center, warehouse, delivery location and required date.
- All monetary values store transaction currency, source exchange-rate reference where used, tax basis and normalized base-currency snapshot for comparison.
- Archive never destroys commercial history. Archived records remain searchable subject to permissions.

8. Procurement Dashboard

The Dashboard is an operational and executive entry point. Every KPI must drill down to the exact underlying records and preserve the user's company/project scope.

KPI group	Required widgets
Demand	Open requests, overdue requests, emergency requests, value awaiting sourcing, requests by project/company/category
Sourcing	Open RFQs, response rate, quotations due, evaluations overdue, average RFQ cycle time, sourcing pipeline
Ordering	Open POs, committed value, unacknowledged POs, overdue PO lines, revised orders, expiring framework capacity
Delivery	Due this week, delayed, partial, rejected, in transit, quantity at risk, critical project deliveries
Suppliers	Qualified, expiring documents, suspended, concentration risk, top performers, performance deterioration
Commercial	Estimated savings, price variance, off-contract spend, budget exposure, committed vs received vs invoiced
My Work	Assigned requests, evaluations, negotiations, orders, delivery exceptions, supplier reviews and linked tasks

PROC-DASH-001

Each dashboard value must state its date basis, currency scope, company/project scope and source records; no unexplained KPI is permitted.

PROC-DASH-002	Dashboard cards must load from cached summaries, while drill-down results use authoritative live data.
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9. Procurement Workspace

The Procurement Workspace is a visual operating board that connects demand to fulfillment. It is not a second source of status; it renders the lifecycle states of underlying records.

Demand	Sourcing	Comparison	Negotiation	Ready to Order	Ordered	Delivery	Completed
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Default procurement pipeline. Stages are configurable through Platform Configuration, but domain transitions remain validated.

- Board, table and timeline views.
- Cards can represent Purchase Requests, Procurement Packages or configured procurement cases.
- Dragging a card proposes a valid transition and opens a confirmation drawer; it must never bypass required fields, locks or future authorization checkpoints.
- Cards show code, title, company/project, value, required date, responsible buyer, current stage, risk indicator and next action.
- Personal Stars, saved views, group-by, filters, column chooser and compact/detailed card modes.
- Exception lane for overdue, missing specification, supplier-document expiry, budget gap, unacknowledged order and delayed delivery.
- Workload indicators by buyer and team without employee-surveillance scoring.

PROC-WORK-001	The board and table must always resolve status from the same backend domain state; no client-only status is allowed.
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PROC-WORK-002	Board transitions must use optimistic concurrency and return a clear conflict message when another user changed the record.
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10. Supplier Management

10.1 Supplier Lifecycle

Prospect	Under Qualification	Qualified	Preferred	Suspended	Blacklisted	Archived
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Suspended and Blacklisted are controlled states. Existing historical records remain visible, but new invitations or orders must be blocked or require an explicit configured override with reason.

10.2 Supplier Master Fields

Field group	Required information
Identity	Supplier code, legal name, trading name, supplier type, registration/tax identifiers, country, addresses, website, language and status.
Contacts	Primary contact, commercial, technical, finance, delivery, warranty and emergency contacts; communication preferences.
Commercial	Currencies, payment terms reference, delivery terms, minimum order values, lead times, price-list references, incoterm readiness, credit notes and negotiated terms.
Categories	Approved categories, products/services, geographic coverage, project size/capacity, brands, manufacturer/dealer status and subcontract trades.
Qualification	Technical capability, quality, HSE, insurance, licenses, certifications, references, capacity, conflicts and qualification decisions.
Finance-sensitive	Bank-account references, tax status and payment hold flags; visibility must be field-permission-ready.
Documents	Registrations, licenses, certificates, insurance, policies, catalogs, price lists, quality records and supplier forms.
Performance	Delivery, quality, commercial, responsiveness, compliance, warranty/after-sales and overall score.
Risk	Concentration, single-source, document expiry, dispute, quality trend, financial/manual risk flag and blacklist reason.
Relationships	Projects, categories, RFQs, quotations, orders, contracts, deliveries, receipts, returns, claims, invoices and payments.

10.3 Supplier Page

- Header: logo, legal/trading name, code, status, supplier type, location, categories, overall score, relationship owner and primary actions.
- Overview: identity, qualification summary, commercial summary, current risks and activity.
- Contacts, Categories & Catalog, Qualification, Documents & Expiry, RFQs & Quotations, Orders, Deliveries, Quality & Returns, Contracts, Finance Summary, Performance, Risks, Communication Timeline and Settings.
- Finance and Contract sections display source-linked summaries only and must respect linked-module permissions.
- Supplier documents use Document Passport and revision history; one file can be referenced in Supplier, Procurement, Contract and Project folders without duplication.

10.4 Qualification and Onboarding

- Configurable qualification templates by supplier type, category, geography, company and project risk class.
- Mandatory/optional questions, required documents, expiration dates, review notes, evidence and outcome.
- Outcomes: Qualified, Conditionally Qualified, Rejected, Suspended, Renewal Required.
- Qualification status can be category-specific and company-specific; one global flag is insufficient.
- Automatic expiration indicators and future notification events.
- No external supplier portal is required in the first release; authorized users enter or import information internally.

SUP-001	A supplier cannot be merged if either record has active commercial documents until duplicate analysis and link migration are completed transactionally.
SUP-002	Blacklisting must preserve reason, scope, actor, date, evidence and review date; historical orders and documents remain intact.

11. Purchase Demand and Purchase Requests

11.1 Request Types

- Stock item / replenishment
- Construction material
- Asset acquisition
- Service
- Subcontract work
- Rental or lease
- Non-stock expense
- Emergency purchase
- Framework call-off request
- Mixed request

11.2 Purchase Request Lifecycle

Draft	Submitted	Under Review	Sourcing	Partially Ordered	Fully Ordered	Closed
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Alternative terminal states: Rejected, Cancelled and Archived. Authorization checkpoints are configurable hooks and may be disabled until the central Approvals Engine is delivered.

11.3 Header Fields

Field group	Fields
Identity	Request code, title, type, priority, status, requester, responsible buyer and creation source.
Organization	Owning company, department, team, project, site, building/structure, unit, work package, cost center and budget line.
Need	Business justification, required-by date, delivery location, warehouse, preferred fulfillment method and urgency reason.
Commercial estimate	Estimated amount, currency, estimate source, available budget summary and requested commitment class.
Sourcing	Preferred/excluded suppliers, single-source reason, existing framework reference, procurement category and suggested package.
Evidence	Specifications, drawings, BOQ, photos, scope, sample requirements and linked Documents.
Control	Confidentiality level, authorization checkpoint status, duplicate warning, risk flags and activity timeline.

11.4 Line Fields

- Line type, item/product/asset/service reference, free-text description and specification revision.
- Quantity, unit, acceptable tolerance, required date and delivery sequence.
- Project, structure, floor, unit, work package, cost code, warehouse/location and requesting team.
- Estimated unit/total cost, currency, tax expectation, freight/landed-cost expectation and budget line.
- Preferred brand/model, allowed alternatives, sample/submittal requirement, quality inspection requirement and warranty expectation.
- Line split across suppliers, dates, locations or orders.
- Attachments and linked task/document references.

11.5 Request Creation Sources

- Manual request from any authorized platform context.
- Inventory reorder recommendation or shortage analysis.
- Asset maintenance work order, replacement or acquisition need.
- Project work package, BOQ activity, site report or material forecast.
- Task, Contract obligation, Unit package, Department need or recurring procurement template.
- Imported request batch with preview, validation and error report.

PR-001

A submitted request must preserve the submitted snapshot. Subsequent edits require return to Draft, a controlled revision or a change note according to configured policy.

PR-002

Duplicate detection must compare company, project, item/specification, required date, quantity, requester and open sourcing/order links before submission.

PR-003

Emergency requests require an emergency reason, risk statement and post-event audit flag even when authorization checkpoints are disabled.

12. Procurement Packages

ConstructionOS differentiator

A Procurement Package is a governed commercial workspace that groups related demand, scope, suppliers, sourcing rounds, purchase orders, contracts, deliveries, documents and costs around one coordinated procurement outcome.

12.1 Package Examples

- Structural concrete package
- Facade package
- Tower A electrical package
- HVAC package
- Interior finishing package
- Elevator package
- Site logistics package
- Office furniture package
- Fleet replacement program
- Annual IT procurement package

12.2 Package Fields

Area	Requirements
Identity	Package code, name, type, status, owning company, project, phase, structures, responsible buyer and technical owner.
Scope	Scope narrative, BOQ/work package references, inclusions, exclusions, technical documents, quantities and package boundaries.
Commercial	Budget reference, estimate, target value, quoted value, awarded value, committed value, received value and variance.
Schedule	Sourcing start, award target, design/submittal dates, manufacture, shipment, phased delivery, installation or service dates.
Suppliers	Longlist, shortlist, invited, responded, technically compliant, negotiated, selected and reserve suppliers.
Records	Linked requests, RFQs, quotations, comparisons, recommendations, orders, contracts, deliveries, receipts, claims and tasks.
Risk	Long-lead risk, single-source, currency, logistics, specification maturity, project critical path and contingency.
Progress	Planning, Sourcing, Awarding, Ordered, Delivery, Completed, Closed, On Hold or Cancelled.

12.3 Package Workspace

- Overview, Scope & BOQ, Requests, Supplier Strategy, RFQs, Quotations, Comparison, Orders, Contracts, Delivery Plan, Budget & Commitments, Documents, Risks, Tasks and Activity.
- Package timeline combines procurement milestones and linked delivery commitments without replacing Project schedule ownership.
- One request line may be split across packages only with explicit quantity allocation; allocated quantity cannot exceed requested quantity.
- One package can produce multiple awards and POs, including phased, geographic or supplier splits.
- Closing a package requires reconciliation of requests, open quantities, orders, returns, claims and linked contracts, or a documented override.

PKG-001

Package totals must be reproducible from linked request, quote, award and PO lines; header totals may be cached but never manually diverge from authoritative lines.

13. RFQs and Tendering

13.1 RFQ Types

- Request for Quotation
- Request for Proposal
- Subcontract tender
- Framework sourcing event
- Single-source negotiation
- Budgetary quotation
- Best and Final Offer round
- Information request linked to later RFQ

13.2 RFQ Lifecycle

Draft	Issued	Open	Responses Received	Evaluation	Negotiation	Awarded	Closed
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Alternative states: On Hold, Cancelled and Archived.

13.3 RFQ Page

- Header with code, title, type, company/project/package, responsible buyer, deadline, status, response count and primary actions.
- Scope and instructions, line schedule, technical documents, commercial form, supplier list, clarifications, response tracking, evaluation criteria, negotiation rounds, recommendation and activity.
- Configurable response fields: price, currency, tax, discount, freight, lead time, delivery dates, payment terms, validity, warranty, country of origin, brand/model, exclusions, alternatives and notes.
- Technical and commercial evaluation sections can be separated and permission-scoped later.
- RFQ issue creates an immutable issued snapshot. Changes after issue create a new RFQ revision or round and preserve supplier response context.

13.4 Supplier Invitations and Responses

- Manual supplier selection, category-based shortlist and previous-performance context.
- Invitation status: Planned, Invited, Viewed/Contacted manually, Intends to Bid, Declined, Response Received, Late, Disqualified.
- Decline reason and no-bid reason capture.
- Multiple quotations per supplier only through explicit revision or sourcing round.
- Initial release supports internal response capture and document upload; supplier portal and secure electronic bidding are future scope.
- Commercial confidentiality and bid-opening permissions must be architecturally supported even if the initial access phase allows all authenticated users.

RFQ-001

An RFQ cannot be issued without at least one line, one supplier, a response deadline and required commercial fields.

RFQ-002

Issued RFQ lines, quantities and specification revisions must be immutable within that issued revision.

RFQ-003

Clarifications must be logged with question, response, audience, documents, actor and timestamp; private supplier clarifications and shared clarifications must remain distinct.

14. Quotations, Comparison and Negotiation

14.1 Quotation Record

Field group	Required data
Identity	Quotation code, supplier reference, supplier quotation number, RFQ/round, received date, validity date, status and source.
Commercial	Currency, exchange-rate snapshot reference, taxes, discount, freight, duties, handling, payment terms and total.
Lines	Offered item/service, description, quantity, unit, unit price, tax, total, lead time, delivery schedule, brand/model, origin and substitute.
Technical	Compliance status, deviations, exclusions, data sheets, samples, method statements, capacity and evaluator notes.
Risk	Validity risk, currency risk, lead-time risk, supplier-document risk, concentration risk and manual risk notes.
Evidence	Original supplier document, revisions, correspondence log and linked Documents.
Outcome	Validated, Clarification Required, Technically Non-Compliant, Commercially Non-Compliant, Shortlisted, Selected, Not Selected, Expired or Withdrawn.

14.2 Comparison Workspace

- Side-by-side supplier columns with sticky line descriptions and normalized totals.
- Separate Technical Evaluation and Commercial Evaluation views.
- Normalization for currencies, tax inclusion/exclusion, freight, duty, discounts, quantities, units, alternates and delivery schedules.
- Total evaluated cost must show the components used; no opaque score is permitted.
- Compare base offer, compliant offer, negotiated offer and final offer.
- Highlight missing lines, quantity differences, unit conversions, exceptions, delivery gaps and commercial exclusions.
- Allow line-by-line award, multi-supplier split, alternate selection and reserve supplier selection.
- Export a comparison report to PDF/Excel when enabled.

14.3 Evaluation Templates

Evaluation dimension	Example criteria
Technical compliance	Specification compliance, approved brand/model, performance, method, sample, capacity and exclusions.
Commercial	Evaluated cost, payment terms, validity, price certainty, discounts, taxes, freight and escalation terms.
Delivery	Lead time, phased dates, logistics plan, location coverage and schedule reliability.
Quality	Certificates, quality plan, inspection readiness, historical rejection/rework and warranty.
HSE / Compliance	Licenses, insurance, HSE evidence, training, required declarations and document validity.
Supplier performance	On-time delivery, quality acceptance, responsiveness, claims, warranty recovery and recent trend.

14.4 Negotiation and Recommendation

- Negotiation round number, date, participants, agenda, supplier position, buyer position, agreed changes, open points and resulting quotation revision.
- Best and Final Offer flag and deadline.
- Award Recommendation records selected supplier(s), selected lines, total evaluated value, reasons, risks, rejected alternatives and supporting evidence.
- Recommendation can be returned for clarification or revised without deleting prior versions.
- Authorization request payload must include recommendation snapshot, comparison, selected suppliers, total, budget reference, risks and documents.

CMP-001	Currency normalization must use a Finance-provided exchange-rate reference and preserve both original and normalized values.
CMP-002	Weighted scores must store every criterion, weight, evaluator input and calculation. Manual overrides require reason and audit history.
AWD-001	Awarding lines must use a transaction lock so the same requested quantity cannot be committed twice through concurrent awards.

15. Purchase Orders and Commercial Control

15.1 Purchase Order Lifecycle

Draft	Issued	Acknowledged	Partially Fulfilled	Fulfilled	Closed
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Additional states: On Hold, Cancelled, Superseded and Archived.

15.2 Purchase Order Header

Field group	Fields
Identity	PO code, title, type, version, status, issue date, supplier order reference and originating award.
Organization	Owning company, branch, department, buyer, project, site, warehouse and cost center.
Supplier	Supplier, order contact, invoice contact, delivery contact and supplier acknowledgment.
Commercial	Currency, subtotal, discounts, taxes, freight, landed-cost estimates, total, payment terms and Finance commitment reference.
Delivery	Delivery address, incoterm readiness, delivery method, requested/confirmed dates, phased schedule and receiving requirements.
Legal	Linked Contract or Framework Agreement, terms reference, warranty, retention readiness and insurance requirements.
Control	Authorization checkpoint reference, confidentiality, attachments, notes, risks, acknowledgment and activity.
Integration	Purchase Request, Package, RFQ, Quotation, Comparison, Inventory expectation, Asset draft, Finance, Documents and Tasks.

15.3 PO Lines

- Line number, source request/award line, type, product/service/asset/subcontract reference, description and specification revision.
- Ordered quantity, unit, unit price, discount, tax, line total, tolerance and currency.
- Requested, promised and revised delivery dates; delivery location and schedule split.
- Project, structure, unit, work package, cost center, warehouse and budget line.
- Inspection, documentation, serial/batch, warranty, sample, submittal and handover requirements.
- Ordered, delivered, accepted, rejected, returned, invoiced and open quantities displayed from source-linked records.

15.4 Issuance, Acknowledgment and Versioning

- Draft PO may be edited. Issued PO is immutable.
- Issue action creates a signed-off internal snapshot, generated PDF representation and Document reference where enabled.
- Supplier acknowledgment captures acknowledged date, supplier reference, accepted/rejected status, confirmed dates and deviations.
- Any commercial or scope change after issue creates a PO Revision with delta amount, delta quantity, date impact, reason and exact changed fields.
- Previous versions remain visible and linked to deliveries, receipts, invoices and claims that referenced them.
- Cancellation may apply to the full PO or remaining open quantity; received/accepted history cannot be erased.

15.5 Framework and Call-Off Ordering

- Framework Agreement legal terms remain in Contracts.
- Procurement stores framework link, approved supplier/category, ceiling value, effective dates, committed/spent value, remaining capacity and call-off rules.
- Call-Off Orders use standard PO lifecycle and exact framework reference.
- Warnings for expired agreement, exceeded ceiling, unapproved category or supplier suspension.
- No automatic legal interpretation; the Contract record remains authoritative.

PO-001	PO totals must reconcile from lines, discounts, taxes, freight and configured charges before issue.
PO-002	Issued POs cannot be edited or deleted. Changes require a revision, remaining-quantity cancellation or superseding order.
PO-003	A PO must store the exact supplier quotation/revision and comparison recommendation from which each awarded line originated.
PO-004	Commitment events must be idempotent so Finance never creates duplicate commitments from retries.

16. Delivery, Receiving, Returns and Claims

16.1 Delivery Schedule

- One PO line may have multiple delivery schedule lines.
- Fields: scheduled quantity, requested date, supplier-confirmed date, dispatch date, expected arrival, actual arrival, location, carrier reference, status and notes.
- Statuses: Planned, Confirmed, Dispatched, In Transit, Delayed, Arrived, Partially Accepted, Accepted, Rejected, Closed.
- Critical project delivery flag and project milestone/work package link.
- Delivery variance reason, revised date, impact note and responsible party.

16.2 Receiving Boundary

Receiving ownership

Procurement owns the expected delivery and supplier commitment. Inventory owns goods receipt, accepted stock quantity, batch/serial capture, warehouse location and posted stock movements. Procurement links those authoritative receipts and shows operational summaries.

- Expected Receipt created from PO delivery schedule.
- Inventory receives stock and returns InventoryReceipt ID, line quantities, accepted/rejected status, inspection state and movement references.
- For asset lines, accepted receipt may create a Draft Asset with serial, photos, warranty and PO linkage; Assets owns activation.
- For non-stock expenses, delivery confirmation may be recorded without Inventory movement.
- For services/subcontracts, use Service Receipt or Work Confirmation rather than goods receipt.

16.3 Delivery Exceptions

- Short delivery
- Over delivery
- Wrong item/specification
- Damaged
- Quality hold
- Missing documents
- Late delivery
- Split delivery
- Rejected delivery
- Unplanned substitution
- Price discrepancy
- Packaging or labeling issue

16.4 Returns and Supplier Claims

- Return record links supplier, PO, delivery, receipt, item/serial/batch, quantity, condition, reason and disposition.
- Return statuses: Draft, Authorized, Dispatched, Received by Supplier, Replacement Pending, Credit Pending, Resolved, Closed.
- Claim types: shortage, damage, quality, warranty, delay, price discrepancy, service failure and documentation failure.
- Claim records store value, evidence, responsible owner, supplier response, resolution, replacement, credit/debit note reference and performance impact.
- Finance owns credit/debit notes and accounting; Procurement links the resulting Finance records.

16.5 Matching Readiness

Procurement provides matching context, while Finance owns the AP invoice and final three-way matching/posting result.

Match source	Authoritative values
Purchase Order	Ordered quantity, price, tax/discount terms, supplier, currency and approved revision.
Inventory / Service Receipt	Accepted quantity, rejected quantity, receipt date, quality state and service acceptance.
Finance Invoice	Invoiced quantity, price, tax, freight, credit/debit adjustments and payment state.
Tolerance configuration	Quantity, price, tax, freight and date tolerances by company/category/supplier/project.

DEL-001	Procurement must never post stock by itself. Goods-receipt summaries are read models backed by Inventory receipt records.
DEL-002	Accepted quantity above ordered tolerance must be blocked or flagged according to configuration and must not silently increase PO quantity.
RET-001	A return must preserve original receipt and movement links; it cannot rewrite the historical accepted receipt.

17. Services and Subcontract Procurement

Construction procurement includes services and subcontract scopes that cannot be modeled as stock. The module must support them natively rather than forcing fake inventory items.

17.1 Service and Subcontract Line Types

- Lump-sum service
- Time and materials
- Rate schedule
- Measured work / BOQ
- Milestone-based service
- Recurring service
- Rental with time/usage
- Subcontract work package
- Professional consultant service
- Transport/logistics service

17.2 Subcontract Tender Requirements

- Scope of work, BOQ/rate schedule, drawings/specifications, inclusions, exclusions, temporary works, site constraints and handover requirements.
- Technical capability, manpower, equipment, program, method statement, HSE, quality plan, insurances, licenses and references.
- Commercial comparison by rates, quantities, preliminaries, provisional sums, taxes, retention readiness, payment terms and exclusions.
- After award, create/link a Contract record for the formal subcontract where required. Procurement retains sourcing and award history.

17.3 Service Receipt / Work Confirmation

Draft	Submitted	Verified	Accepted	Invoice Eligible	Closed
-------	-----------	----------	----------	------------------	--------

- PO/contract line, service period, milestone, work package, measured quantity, rate, value and completion percentage.
- Witness/evaluator, evidence photos, report, timesheet, measurement sheet, deficiencies and withheld quantity/value.
- Accepted value and quantity become matching context for Finance AP.
- Defects or incomplete work can create linked Tasks, Claims or future QA/QC records.

SRV-001

Service acceptance must be distinct from PO issuance and supplier invoice. No service invoice should be treated as accepted work solely because it exists in Finance.

18. Cross-Module Integrations

Module	Integration contract
Projects	Read project hierarchy, work packages, cost codes, milestones, sites and delivery context. Show procurement status in Project Page without duplicating records.
Inventory	Read product/UoM and reorder signals. Send expected receipts. Link authoritative receipts, rejections, batches/serials and supplier returns.
Assets	Create/link Draft Asset from accepted asset purchase. Pass acquisition, serial, warranty, supplier, PO and receipt context.
Finance	Read budgets, currencies, exchange rates, tax references and AP states. Send commitment-ready PO events. Link invoices, matching, payments and price variance.
Contracts	Initiate purchase/subcontract/framework contract after award. Link exact Contract and call-off capacity; never duplicate signed clauses.
Documents	Generate procurement system folders. Store RFQ, quotes, comparisons, POs, delivery notes and supplier documents as Document records with revisions.
Tasks	Create/link participant-based tasks for specification, evaluation, clarification, delivery exception, claim and supplier review.
Teams / Users	Assign requester, buyer, technical evaluator, package owner, delivery coordinator and observers; no automatic broad task exposure.
Calendar	Expose RFQ deadlines, quotation expiry, delivery dates, supplier-document expiry and package milestones as future calendar events.
Notifications	Emit events for deadlines, exceptions and assigned work; notification preferences and delivery are owned by the future Notifications module.
Approvals	Expose authorization checkpoints and immutable request payloads; routing, delegation, escalation and decision state are owned by the final Approvals PRD.

18.1 Document Folder Architecture

- Procurement root system folder per company and optional project shortcuts.
- 01 Purchase Requests
- 02 Procurement Packages
- 03 RFQs and Tender Documents
- 04 Supplier Quotations
- 05 Comparisons and Recommendations
- 06 Purchase Orders
- 07 Deliveries and Receipts
- 08 Supplier Documents
- 09 Contracts and Frameworks
- 10 Returns and Claims
- 11 Finance References
- Transzit

Documents may appear by shortcut in Supplier, Project, Asset, Inventory, Contract and Finance folders. The underlying Document and file must not be duplicated.

19. Search, Views, Reporting and Configuration

19.1 Search

- Global search by code, title, supplier, supplier quotation number, PO reference, product/SKU, project, package, work package, requester, buyer, delivery location and document metadata.
- Results respect tenant, company, project, record, field and confidentiality policies.
- Search index stores safe display fields only; sensitive commercial details are fetched after authorization.
- Recent searches, personal Stars and saved views.

19.2 Filters and Saved Views

- Company, project, department, buyer, requester, supplier, category, type, status, priority, required date, RFQ deadline, delivery date, currency, value range, budget status, risk, package, framework, document expiry and performance band.
- Saved view stores page, layout, filters, grouping, sorting, columns, density and selected KPI context.
- Saved views are private by default; shared views are a later capability.
- Disabled Platform Configuration elements are automatically removed from saved views without leaving broken filters.

19.3 Reports and Analytics

Report family	Required reports / metrics
Demand	Request aging, emergency demand, request-to-order cycle, demand by project/category/company and unfulfilled demand.
Spend	PO spend, committed spend, received value, supplier/category/project spend, currency exposure and spend trend.
Savings	Baseline vs selected evaluated cost, negotiated savings, cost avoidance, price variance and savings methodology.
Supplier	On-time delivery, fill rate, quality acceptance, response rate, claim rate, lead-time accuracy, warranty recovery and score trend.
Sourcing	RFQ response rate, competition level, cycle time, technical compliance, quotation validity and award distribution.
Order	Open PO value, overdue lines, acknowledgment rate, revision frequency, cancellation, framework usage and off-contract spend.
Delivery	Late delivery, partial delivery, reject rate, quantity variance, critical deliveries and project impact.
Risk	Supplier concentration, single-source exposure, expiring documents, suspended supplier exposure and long-lead packages.
Audit	Lifecycle transitions, overrides, revisions, imports, exports, sensitive changes and user activity.

19.4 KPI Formula Transparency

- Every KPI displays definition, numerator, denominator, exclusions, date basis, currency conversion method and source records.
- Estimated savings must identify baseline type: budget, previous price, initial quote, market estimate or negotiated starting point.
- On-time delivery uses the configured committed date and exact accepted receipt or service acceptance date.
- Supplier score weights are configurable and versioned; historic scores retain the weight set used at calculation time.

19.5 Platform Configuration

- Module, page, section, tab, widget, field group, feature, action, lifecycle branch, line type, report, export and authorization checkpoint enable/disable.
- Configurable supplier types, categories, request types, RFQ types, evaluation templates, scoring weights, numbering, currencies, tolerances, units, statuses and reason codes.
- Disabled features disappear cleanly from navigation, global search, API discovery, page layout and saved views.
- Configuration and permissions are evaluated separately: enabled functionality may still be unauthorized for a user.

CFG-001

Platform Configuration changes must be versioned and auditable. Historic records continue to render using preserved values even when a type or field is later disabled.

20. Responsive UX, Premium Motion and Accessibility

20.1 Desktop

- Wide data tables with sticky identifiers, column chooser, resizable columns, virtualized rows and drill-down drawers.
- Comparison Workspace supports horizontally scrollable supplier columns with frozen scope and totals.
- Procurement Package and Supplier Page use compact tab/section navigation with lazy-loaded heavy sections.
- Right-side preview panels preserve list context for requests, quotations, POs and documents.

20.2 Tablet

- Two-column dashboards
- Collapsible navigation
- Horizontal comparison/table scroll when unavoidable
- Touch-safe actions
- Bottom-sheet filters and record actions

20.3 Mobile Web

- Create or update a request, attach photos/documents, open supplier/PO, confirm delivery arrival, record shortage/damage, capture service receipt and create a return/claim.
- Compact sticky record header with code, status and primary action menu.
- Comparison becomes supplier-by-supplier cards with a persistent line selector; no unreadable desktop table shrink.
- Camera upload and QR/barcode readiness for PO/delivery lookup. Full offline synchronization is future scope.

20.4 Premium Motion

- Subtle card lift and shadow on hover; 140-220 ms transitions.
- Animated KPI value transitions only when data changes and without delaying comprehension.
- Smooth board movement followed by server confirmation; revert animation on validation conflict.
- Skeleton loading for dashboards, lists, comparison and activity.
- Quick actions reveal on row hover without shifting layout.
- Status transitions use restrained semantic motion; no celebratory animation for high-risk commercial actions.
- Reduced-motion preference removes non-essential motion while preserving state feedback.

20.5 Accessibility

- WCAG 2.2 AA target, keyboard navigation, visible focus, semantic headings, accessible tables and labeled controls.
- Color never acts as the only status indicator.
- Comparison tables expose headers and relationships for assistive technology.
- All drag actions have keyboard and menu alternatives.
- Charts provide textual summaries and underlying data tables.
- Minimum touch target and responsive zoom support.

21. Data Model and Storage

21.1 Core Entities

Entity	Purpose
Supplier	Purchasing profile linked to optional shared business-party identity.
SupplierContact	Role-based supplier contact.
SupplierCategoryAssignment	Approved category and company/project scope.
SupplierQualification	Template result, evidence, outcome and validity.
SupplierDocumentLink	Link to Document record and required-document type.
SupplierScorecard	Versioned performance period and metric results.
SupplierRiskFlag	Manual or calculated risk with scope and lifecycle.
PurchaseRequest	Demand header.
PurchaseRequestLine	Demand line with item/service/project dimensions.
ProcurementPackage	Coordinated commercial package.
ProcurementPackageLink	Links requests, RFQs, POs, contracts and deliveries.
RFQ	Sourcing event header and issued revision.
RFQLine	Requested line and response schema.
RFQSupplier	Invitation and response state per supplier.
RFQClarification	Question/response with audience scope.
Quotation	Supplier offer header and revision.
QuotationLine	Offered line, terms, compliance and normalization.
EvaluationTemplate	Versioned criteria and weights.
QuotationEvaluation	Evaluator inputs and calculated result.
BidComparison	Comparison snapshot and normalization rules.
NegotiationRound	Commercial/technical negotiation history.
AwardRecommendation	Versioned selected supplier/line recommendation.
PurchaseOrder	Order header and current lifecycle.
PurchaseOrderLine	Committed commercial line.
PurchaseOrderRevision	Immutable delta and issued revision snapshot.
DeliverySchedule	Planned fulfillment segment.
Delivery	Supplier shipment/arrival record.
DeliveryLine	PO schedule quantity and exception context.
ServiceReceipt	Accepted service/work measurement.
SupplierReturn	Return to supplier and resolution.
SupplierClaim	Commercial/quality/delivery claim.
FrameworkUsage	Call-off and ceiling consumption reference.
ProcurementCommitmentLink	Finance commitment reference and idempotency key.
ProcurementActivity	Immutable domain activity.
ProcurementSavedView	Private user view configuration.
ProcurementNumberSequence	Configurable code generator.

21.2 Common Technical Fields

```
id UUID, tenant_id UUID, company_id UUID, project_id UUID NULL,
code VARCHAR, status VARCHAR, version INTEGER,
```

```
created_at TIMESTAMPTZ, created_by UUID, updated_at TIMESTAMPTZ, updated_by UUID,
archived_at TIMESTAMPTZ NULL, correlation_id UUID,
configuration_snapshot JSONB, custom_fields JSONB, row_version BIGINT
```

21.3 Data Integrity Principles

- Foreign keys and scope validation for tenant/company/project compatibility.
- Soft archive for master and commercial records; no destructive cascade across modules.
- Immutable issued snapshots stored separately from editable current drafts.
- Monetary decimals use fixed precision; quantity precision is unit-configurable.
- Domain events written through an outbox in the same transaction as the source change.
- Activity records are append-only and cannot be edited by application users.
- Custom fields are schema-validated through Platform Configuration and indexed selectively.

22. API Contract

The API is REST-first with event-driven integration. Endpoint examples are version-neutral; the implementation must follow the platform API conventions for authentication, pagination, errors, idempotency, filtering and audit headers.

22.1 Core Endpoints

Endpoint	Purpose
GET /procurement/dashboard	Scoped KPI and work summaries.
GET /procurement/workspace	Board/table/timeline query.
GET /suppliers	Search/filter supplier directory.
POST /suppliers	Create supplier.
GET /suppliers/{id}	Supplier Page aggregate.
PATCH /suppliers/{id}	Update editable supplier fields.
POST /suppliers/{id}/qualifications	Create qualification assessment.
POST /suppliers/{id}/status-transitions	Qualified/preferred/suspended/blacklisted transition.
GET /purchase-requests	List requests.
POST /purchase-requests	Create request.
POST /purchase-requests/{id}/submit	Submit immutable snapshot.
POST /purchase-requests/{id}/revise	Create controlled revision.
POST /purchase-requests/{id}/convert-to-rfq	Create RFQ from selected lines.
GET /procurement-packages	List packages.
POST /procurement-packages	Create package.
POST /procurement-packages/{id}/allocate-lines	Allocate request quantities.
POST /rfqs	Create RFQ.
POST /rfqs/{id}/issue	Issue exact revision.
POST /rfqs/{id}/suppliers	Add supplier invitation.
POST /rfqs/{id}/clarifications	Create clarification.
POST /rfqs/{id}/quotations	Capture supplier quotation.
POST /comparisons	Create comparison snapshot.
POST /comparisons/{id}/evaluate	Store evaluator inputs.
POST /comparisons/{id}/recommend	Create award recommendation.
POST /purchase-orders	Create PO from award or direct request.
POST /purchase-orders/{id}/issue	Issue immutable PO revision.
POST /purchase-orders/{id}/acknowledge	Capture supplier acknowledgment.
POST /purchase-orders/{id}/revisions	Create PO revision.
POST /purchase-orders/{id}/cancel-remaining	Cancel open quantity.

Endpoint	Purpose
POST /deliveries	Create/record delivery.
POST /deliveries/{id}/exceptions	Record delivery exception.
POST /service-receipts	Create service/work receipt.
POST /supplier-returns	Create return.
POST /supplier-claims	Create claim.
GET /procurement/reports/{reportKey}	Generate/report query.
GET /procurement/settings	Read configuration.
PATCH /procurement/settings	Update configuration with audit.

22.2 API Standards

- Cursor pagination for large lists and activity feeds.
- Filter syntax supports company/project scope, status, date, supplier, category, value and saved-view state.
- Idempotency-Key required for issue, conversion, commitment, receipt-link and bulk actions.
- If-Match or row_version required for updates and transitions.
- Consistent error model: code, message, field, correlation ID, retryable flag and remediation hint.
- Bulk endpoints use asynchronous jobs with validation preview and downloadable error report.
- Exports and heavy reports run asynchronously and return job status.

22.3 Example Domain Payload

```
{
  "event": "purchase_order.issued",
  "tenantId": "...", "companyId": "...", "purchaseOrderId": "...",
  "purchaseOrderVersion": 3, "supplierId": "...",
  "currency": "EUR", "total": "125000.00",
  "projectDimensions": [{"projectId": "...", "workPackageId": "..."}],
  "authorizationCheckpointRef": null,
  "actorId": "...", "correlationId": "...", "occurredAt": "..."
}
```

23. Backend Services, Events and Jobs

23.1 Domain Services

- Supplier master and qualification service
- Supplier scoring and risk service
- Purchase request service
- Procurement package service
- RFQ and sourcing-round service
- Quotation capture and normalization service
- Evaluation and comparison service
- Negotiation and recommendation service
- Purchase order and revision service
- Delivery and exception service
- Service receipt service
- Return and claim service
- Framework usage service
- Finance commitment integration service
- Inventory receiving integration service
- Asset creation integration service
- Document linking service
- Search indexing service
- Report generation service
- Activity/audit writer
- Platform Configuration resolver
- Permission policy adapter
- Cache invalidation service

23.2 Domain Events

- supplier.created
- supplier.qualification_changed
- supplier.status_changed
- supplier.document_expiring
- supplier.performance_updated
- purchase_request.created
- purchase_request.submitted
- purchase_request.revised
- purchase_request.cancelled
- purchase_request.converted
- procurement_package.created
- procurement_package.status_changed
- rfq.issued
- rfq.deadline_changed
- rfq.response_received
- rfq.closed
- quotation.received
- quotation.revised
- quotation.expiring
- comparison.created
- award.recommended
- award.selected
- purchase_order.issued
- purchase_order.acknowledged
- purchase_order.revised
- purchase_order.cancelled
- procurement.commitment_changed
- delivery.scheduled
- delivery.dispatched
- delivery.delayed
- delivery.arrived
- inventory_receipt.linked
- service_receipt.accepted
- supplier_return.created
- supplier_claim.opened
- supplier_claim.resolved

23.3 Background Jobs

- Supplier document and qualification expiry evaluation.
- RFQ deadline, quote validity and clarification reminder events.
- Unacknowledged PO and overdue delivery evaluation.
- Package milestone and long-lead risk evaluation.
- Supplier performance recalculation from receipts, returns, claims and delivery outcomes.
- Commitment reconciliation with Finance references.
- Search indexing, dashboard aggregation and cache reconciliation.
- Bulk import processing, report generation and export generation.
- Orphan-link detection and cross-module integrity checks.

Background-job rule

All jobs must be idempotent, tenant-aware, correlation-aware, safe to retry and observable. A retry must not issue a second PO, duplicate a Finance commitment or create a second Inventory expectation.

24. Frontend Architecture

24.1 Application Structure

- Route-level modules for dashboard, workspace, suppliers, requests, packages, sourcing, orders, delivery, reports and settings.
- Shared procurement component library: status chip, money display, supplier badge, line table, comparison matrix, timeline, activity feed, document links and risk banner.
- Page shell uses standard ConstructionOS sidebar, global search, filters, notifications and user menu.
- Heavy sections are lazy-loaded and deep-linkable by tab/anchor.
- Feature visibility resolved through Platform Configuration before rendering; action availability resolved through permissions and record state.

24.2 State and Data Fetching

- Server state is authoritative. Client state stores UI preferences, draft form state and optimistic transition intent only.
- Query keys include tenant/company/project scope, filters, sorting and pagination.
- Mutations invalidate exact affected aggregates and apply event-driven incremental counters where available.
- Autosave for long drafts with explicit saved status and recovery; issued records never autosave modifications.
- Form schemas share validation definitions with backend where practical but backend remains authoritative.
- Comparison Workspace uses virtualization for large line/supplier matrices.

24.3 Error and Empty States

- Actionable empty states with Create Request, Add Supplier or Create RFQ according to context.
- Conflict messages state who/what changed and offer Reload or Open New Version.
- Integration delay states distinguish Pending, Failed and Reconciled links.
- Permission denial never reveals hidden values in error messages.
- Disabled configuration sections reflow without blank cards, dead tabs or broken deep links.

25. Validation, Concurrency, Security and Audit

25.1 Validation Rules

- Supplier code unique in configured scope; tax/registration identifier duplicates produce review warning.
- Request lines require positive quantity/value where applicable, valid unit and compatible company/project scope.
- Required date cannot precede request date unless emergency override reason is provided.
- RFQ requires valid deadline, at least one line and at least one supplier.
- Quotation totals reconcile to lines, discounts, taxes and charges.
- Quotation revision cannot silently replace the original supplier document.
- Evaluation weights total 100% when weighted scoring is enabled.
- Awarded quantity cannot exceed open requested quantity unless configured overage with reason.
- PO totals reconcile and issued revision is immutable.
- PO supplier, currency and company must match selected award unless explicit direct-order flow is used.
- Delivery received quantity cannot exceed open PO tolerance without exception.
- Service receipt accepted quantity/value cannot exceed open PO/contract tolerance.
- Return quantity cannot exceed linked accepted quantity net of prior returns.
- Framework call-off cannot exceed remaining ceiling or effective dates without configured override.
- Archive requires impact check for open requests, RFQs, orders, deliveries, claims and cross-module links.

25.2 Concurrency

- Optimistic row version for editable records.
- Transactional quantity allocation locks for award, PO creation, delivery acceptance context and return quantity.
- Idempotency for conversions, issue actions and external-module events.
- Immutable snapshot hash for issued RFQ, quotation import, comparison recommendation and PO revision.
- Bulk operations return per-record success/failure; no silent partial state.

25.3 Security and Permission Readiness

- Tenant isolation and company/project scope checks on every query and mutation.
- Record-level, section-level, field-level and action-level policy hooks.
- Sensitive-field readiness for supplier bank data, quotation pricing, comparison, negotiation notes, risk, conflicts and Finance summaries.
- Commercial bid confidentiality and future sealed-bid readiness.
- No linked-module access expansion.
- Export permission and watermark readiness for confidential commercial reports.
- Sensitive read-audit readiness for quotations, comparisons, bank data and negotiation notes.
- Platform Configuration evaluated separately from permissions.

25.4 Suggested Capability Keys

```
procurement.view, procurement.dashboard.view, procurement.reports.view,
suppliers.view, suppliers.manage, suppliers.qualify, suppliers.suspend, suppliers.view_sensitive,
purchase_requests.create, purchase_requests.submit, purchase_requests.manage,
procurement_packages.manage, rfqs.create, rfqs.issue, quotations.manage,
comparisons.view, comparisons.evaluate, awards.recommend,
purchase_orders.create, purchase_orders.issue, purchase_orders.revise, purchase_orders.cancel,
deliveries.manage, service_receipts.manage, returns.manage, claims.manage,
procurement.export, procurement.settings.manage
```

Current phase policy may return ALLOW for all authenticated users, but every frontend action and backend endpoint must call these hooks.

25.5 Audit and Immutability

- Every lifecycle transition stores actor, timestamp, previous/new status, reason, source, version and correlation ID.
- Every issued RFQ/PO and submitted request stores an immutable snapshot.
- Every quotation revision and supplier document revision remains linked.
- Every comparison score and override stores evaluator and before/after values.
- Every supplier qualification, suspension, blacklist and risk change is immutable in activity history.
- Imports, exports, bulk actions and configuration changes are logged.
- Audit events are append-only and exportable according to permission.

26. Performance, Observability and Testing

26.1 Performance and Scalability

- Cursor pagination and virtualized lists for suppliers, requests, quotations, POs and activity.
- Virtualized comparison matrix supporting hundreds of lines and dozens of suppliers.
- Indexes by tenant, company, project, supplier, category, status, code, dates and source links.
- Precomputed dashboard and supplier-performance summaries updated incrementally by events.
- Asynchronous exports, imports, reports and large comparison calculations.
- No N+1 queries in line aggregates and linked-module summaries.
- Target initial dashboard/standard list interaction below 2 seconds under normal enterprise conditions.
- Target cached drill-down below 250 ms and common search responses below 500 ms.
- Support millions of procurement lines and activity events with partitioning readiness.

26.2 Observability

- API latency, error rate, timeout and retry metrics by endpoint.
- Event-outbox backlog, integration failures and reconciliation state.
- RFQ issue/PO issue success, idempotency collision and transition conflict rate.
- Import success/error rate, report generation duration and search indexing delay.
- Dashboard cache freshness and supplier score calculation duration.
- Business telemetry: request cycle time, response rate, PO acknowledgment and delivery exception rate.
- No employee-surveillance analytics or hidden individual performance scoring.

26.3 Testing Strategy

Test layer	Required coverage
Unit	Lifecycles, totals, currency normalization, evaluation weights, quantity allocation, PO revisions, tolerances, supplier score, numbering and Platform Configuration resolution.
Integration	Projects dimensions, Inventory products/receipts/returns, Assets draft creation, Finance budgets/commitments/AP, Contracts creation/framework, Documents revisions, Tasks privacy and event outbox.
End-to-end	Create supplier; qualify; create request; package; issue RFQ; capture quotes; compare; negotiate; recommend; issue PO; acknowledge; receive partial delivery; reject quantity; create return/claim; close.
Security	Tenant leakage, company/project scope, sensitive commercial fields, exports, linked-module access and unauthorized mutation.
Performance	Large supplier directory, 10k-line requests, multi-supplier comparison, million-line PO history, dashboard concurrency and bulk imports.
Accessibility	Keyboard-only, screen reader, focus order, table semantics, comparison alternatives, reduced motion and mobile zoom.
Visual regression	Dashboard, Workspace board/table, Supplier Page, Request Page, Package Page, RFQ Page, Comparison, PO Page, Delivery, tablet, mobile, long labels and disabled-section reflow.

27. Migration, Implementation Phases and Acceptance

27.1 Migration and Rollout

1. Define core supplier, request, RFQ, quotation, PO and activity tables with tenant/company scope.
2. Map existing suppliers and deduplicate using controlled review; do not auto-merge commercial histories.
3. Import open requests, active RFQs, quotations and POs with source identifiers and migration audit.
4. Link existing Inventory products, Finance suppliers/AP records, Contracts and Documents.
5. Generate Procurement system folders and document shortcuts.
6. Backfill PO open quantities and delivery status from authoritative receipts/invoices where possible.
7. Rebuild search indexes and dashboard aggregates.
8. Pilot with one company/project and reconcile counts, values, statuses and links.
9. Enable additional companies through Platform Configuration and monitor integration events.
10. Freeze or archive legacy procurement records only after signed reconciliation report.

27.2 Implementation Phases

Phase	Scope
1 - Foundation	Supplier master, categories, qualification, numbering, configuration, permissions hooks, documents and search.
2 - Demand	Purchase Requests, request lines, imports, duplicate detection, replenishment/project/asset origins.
3 - Packages and Sourcing	Procurement Packages, RFQs, invitations, quotations, clarifications and tender revisions.
4 - Evaluation and Award	Comparison, normalization, technical/commercial evaluation, negotiation and recommendation.
5 - Ordering	Purchase Orders, issuance, acknowledgment, revisions, call-offs and Finance commitment events.
6 - Delivery and Supplier Control	Delivery schedules, receipt links, service receipts, returns, claims and supplier scorecards.
7 - Analytics and Hardening	Reports, dashboards, performance, accessibility, security, migration, observability and complete testing.
Future final integration	Connect all defined checkpoints to the central Approvals and Workflow Engine after that engine PRD is completed.

27.3 Acceptance Criteria

- AC-01 - The module supports multiple companies, projects, sites, departments and procurement categories.
- AC-02 - Suppliers support qualification, documents, category scope, performance, risk and complete history.
- AC-03 - Purchase Requests support stock, assets, services, subcontract work, rentals and mixed lines.
- AC-04 - Purchase Request revisions and submitted snapshots are preserved.
- AC-05 - Procurement Packages group requests, sourcing, orders, contracts, deliveries and costs.
- AC-06 - RFQs support multiple suppliers, issued revisions, clarifications, deadlines and sourcing rounds.
- AC-07 - Quotations preserve original and revised supplier offers.
- AC-08 - Comparison supports technical/commercial evaluation, normalization, line awards and multi-supplier splits.
- AC-09 - Evaluation scores and overrides are transparent and auditable.
- AC-10 - Purchase Orders support issuance, acknowledgment, revisions, cancellation and immutable history.
- AC-11 - Framework call-offs link to Contracts and track capacity without duplicating legal terms.
- AC-12 - Delivery schedules support partial, delayed, rejected and phased fulfillment.
- AC-13 - Inventory owns authoritative goods receipts and stock movements; Procurement displays linked summaries only.
- AC-14 - Service and subcontract receipts are supported without fake inventory transactions.
- AC-15 - Returns and claims preserve receipt, finance and supplier-performance links.
- AC-16 - Finance commitment, invoice, payment and budget data remain source-linked.
- AC-17 - Asset purchases can create linked Draft Assets without transferring lifecycle ownership.
- AC-18 - Documents use Document Passport, revisions, shortcuts and Transit without duplication.
- AC-19 - Tasks remain participant-based and do not expose procurement records to entire teams automatically.
- AC-20 - Every key KPI has a visible formula, source and scope.
- AC-21 - Search, filters, saved views, Stars, bulk import/export and audit are implemented.
- AC-22 - Desktop, tablet and mobile retain all critical operational actions.
- AC-23 - Premium motion and reduced-motion support are implemented.
- AC-24 - Every module element supports Platform Configuration and disabled features reflow cleanly.
- AC-25 - Every action is permission-hooked and ready for later record/field restrictions.
- AC-26 - Issued commercial records and audit history are immutable.

- AC-27 - Concurrency controls prevent duplicate awards, commitments and over-allocation.
- AC-28 - Background jobs and integration events are idempotent and tenant-aware.
- AC-29 - Large line, supplier and activity volumes remain performant.
- AC-30 - Authorization checkpoints are exposed for future integration without embedding the final workflow engine.

28. Page and Flow Registry

Page / Flow	Page / Flow
Procurement Dashboard	My Procurement Work
Procurement Workspace - Board	Procurement Workspace - Table
Procurement Workspace - Timeline	Supplier Directory
Supplier Page	Create/Edit Supplier
Supplier Qualification	Supplier Document Renewal
Supplier Performance Scorecard	Supplier Suspension/Blacklist
Purchase Requests	Purchase Request Page
Create/Revise/Cancel Request	Request Import
Procurement Packages	Procurement Package Page
Allocate Request Lines	RFQs
RFQ Page	Create/Issue/Revise/Cancel RFQ
Supplier Invitation	RFQ Clarification
Quotation Capture	Quotation Revision
Comparison Workspace	Technical Evaluation
Commercial Evaluation	Negotiation Round
Award Recommendation	Purchase Orders
Purchase Order Page	Create/Issue/Acknowledge PO
PO Revision	Cancel Remaining Quantity
Framework Call-Off	Delivery Schedule
Delivery Page	Delivery Exception
Expected Receipt Link	Service Receipt / Work Confirmation
Supplier Return	Supplier Claim
Procurement Reports	Procurement Import/Export
Procurement Settings	Saved Procurement Views
Procurement Activity and Audit	

29. Future Roadmap

- Central Approvals and Workflow Engine integration with routing, delegation, escalation, thresholds and decision policies.
- Supplier Portal for onboarding, RFQ response, document renewal, order acknowledgment, delivery updates and claims.
- Email, EDI, cXML, PEPPOL/e-invoicing and external ERP integrations.
- Electronic tender opening, sealed bids and e-auctions.
- Digital signatures and external contract-signing providers.
- OCR and structured extraction for supplier quotations, delivery notes and invoices.
- AI-assisted specification comparison, supplier recommendation, anomaly detection and lead-time prediction with human review.
- Live logistics/courier tracking and shipment milestones.
- External supplier-risk, sanctions and credit-data services where legally appropriate.
- Offline mobile draft capture and field delivery workflows.
- Marketplace/catalog integrations and punchout purchasing.
- Advanced category management, demand forecasting and strategic sourcing scenarios.

Future capabilities must remain optional. The first release must be fully usable without third-party integrations or AI services.

30. Final Architectural Rule

Final rule

Procurement owns supplier purchasing context and the controlled journey from demand through sourcing, award, purchase commitment and delivery expectation. Inventory owns stock and receipt movements, Finance owns budgets and accounting, Contracts owns legal agreements, Assets owns asset lifecycle, Projects owns work context, Documents owns files, Tasks owns execution, and the final Approvals Engine will own decision routing. ConstructionOS links these records through stable IDs and events without duplicating authority.

This separation must remain visible in the data model, APIs, UI labels, permissions, audit history and acceptance tests.